

ELEC50001 – Circuits and Systems

SAMPLE PAPER 2

Answer ALL questions.

There are THREE questions on the paper.

Question ONE counts for 50% of the marks, other questions 25% each

Time allowed: 2 hours

Information for Candidates:

The following notation is used in this paper:

1. Unless explicitly indicated otherwise, digital circuits are drawn with their inputs on the left and their outputs on the right.
2. Within a circuit, signals with the same name are connected together even if no connection is shown explicitly.
3. The notation $X[2:0]$ denotes the three-bit number X_2 , X_1 and X_0 . The least significant bit of a binary number is always designated bit 0.

1. *Figure 1.1* shows an inverting amplifier circuit constructed from an operational amplifier with a Gain-Bandwidth product of 1MHz that has a single power supply at 5V, two resistors R1 and R2, and a voltage source at 2.5V.

- (i) Write down an equation for the Gain of this amplifier. [2]
- (ii) Given that the amplifier has a gain of -14 and $R1 = 2.2k\Omega$, determine the value of R2. [2]
- (iii) A design is required to provide an overall gain of -200 for a signal with frequency component from 1kHz to 50kHz. Explain why the circuit in *Figure 1.1* is unable to achieve the specification. [2]
- (iv) Hence or otherwise, design an additional circuit using a second operational amplifier to meet the required specification. [4]

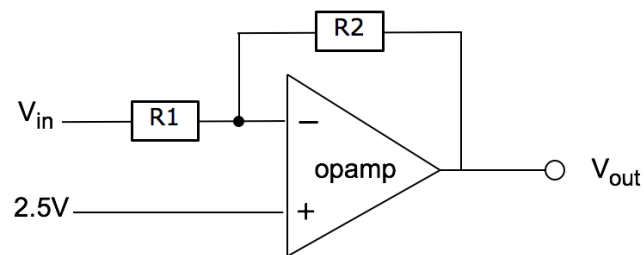


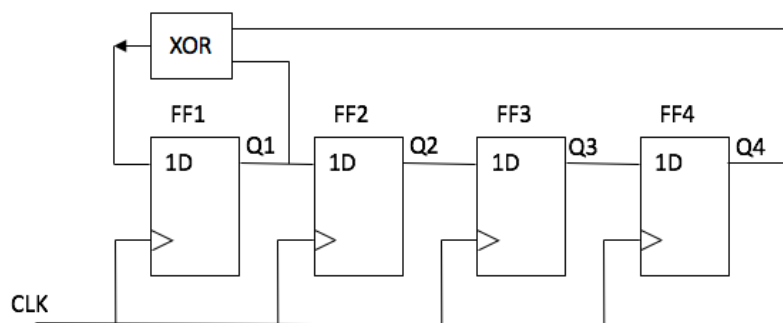
Figure 1.1

(b) *Figure 1.2* depicts a circuit with four D-type flip-flops FF1-FF4 connected in series with an XOR gate.

(i) If FF1, FF2 and FF4 are initially reset, and FF3 is initially set, determine the output Q4:1 over 10 clock cycles. [2]

(ii) Implement this circuit in Verilog HDL with an interface defined in *Figure 1.2*. [5]

(iii) Given that $1 + x^3 + x^5$ is a primitive polynomial, design in the form of a schematic circuit diagram a linear-feedback shift register that produces a 5-bit pseudo-random binary number. [3]



```
module sr4 (
    clk,
    q);
    input clk;
    output [4:1] q;
```

Figure 1.2

(c) Explain the principle of operation of a 12-bit pulse-width modulation (PWM) digital-to-analogue converter (DAC). Design in Verilog HDL a PWM DAC with an interface as defined in *Figure 1.3*.

[10]

```
module pwm_dac (clk, data_in, pwm_out);

    input      clk;           // system clock
    input [11:0] data_in;     // input data for conversion
    output     pwm_out;       // PWM output
```

Figure 1.3

- (d) *Figure 1.4* shows a circuit with two D-flipflops FF1 and FF2 with setup and hold times of 3 ns and 2 ns respectively, and a clock-to-Q output delay of 2 ns. The clock signal CLK has a 1:1 mark-space ratio. The datapath is driven by logic_A, which has a propagation delay between 2 ns and 5 ns. The clock path is driven by logic_B, which has a propagation delay between 1 ns and 10 ns. Derive the maximum frequency of the signal CLK for reliable operation of this circuit.

[10]

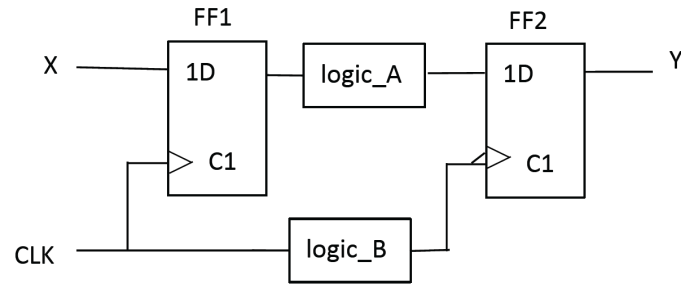


Figure 1.4

(e) *Figure 1.5* shows a counter circuit where one of the flip-flops has an asynchronous reset input which forces Q2 to zero whenever M=1.

(i) For M=0, draw the state diagram for the circuit of *Figure 1.5* assuming that all flip-flops are initially in the reset state. Draw a timing diagram showing the waveforms of the clock CLK, and the outputs Q0, Q1 and Q2 for one complete cycle of the counter.

[6]

(ii) Repeat (i) above, but for the case where M=1. By considering the output Q0, state the behaviour of this counter in relation to the control signal M.

[4]

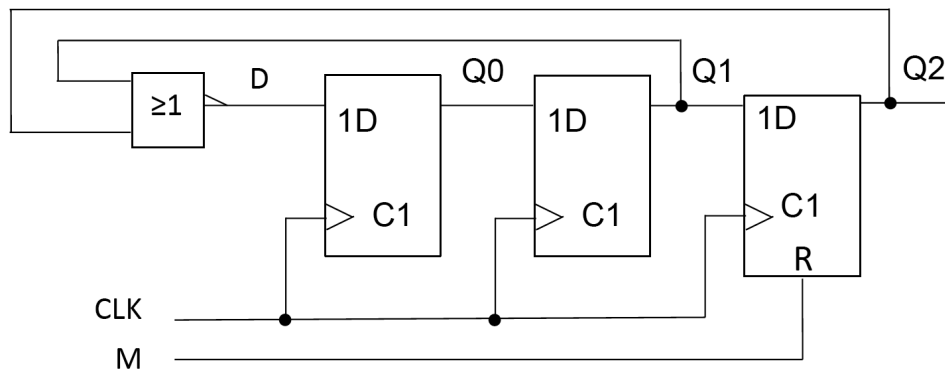


Figure 1.5

1. *Figure 2.1* shows a finite state machine (FSM) with 1-bit input IN and 2-bit output OUT1:0, implemented using an 8 x 4-bit ROM and a 4-bit register. The input IN to the FSM drives the upper bit of the ROM address A2. The output of the ROM D3:0 is connected to the data input of the register. The lower 2 bits of the ROM data D1:0 provide the next state value, which are fed back to the lower 2 bits of the ROM address A1:0 through the state register S1:0. The upper 2 bits of the ROM data D3:2 provide the FSM output value. The reset input RST clears the register.

Figure 2.2 shows the state diagram of the FSM.

- (a) Determine the contents of the ROM in order to implement this FSM. [8]
- (b) A Logic Element (LE) in Intel' FPGA device consists of a 4-input lookup table and a register. Design the SSM using LEs in the Cyclone II device using *one-hot state encoding*. Estimate how many LEs are required to implement the FSM. [10]
- (c) Write a synthesizable Verilog module to describe the FSM in (b). [7]

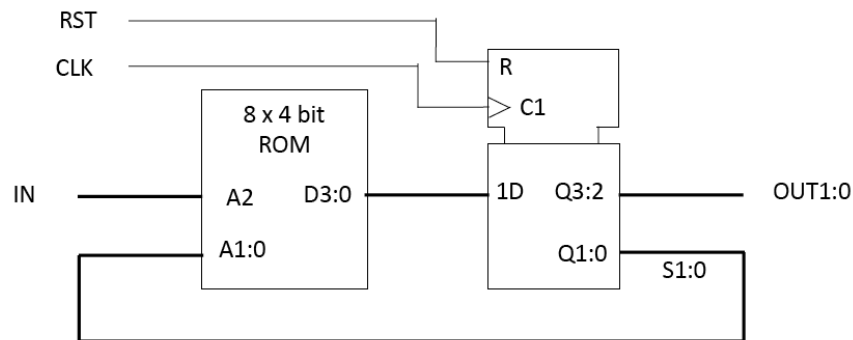


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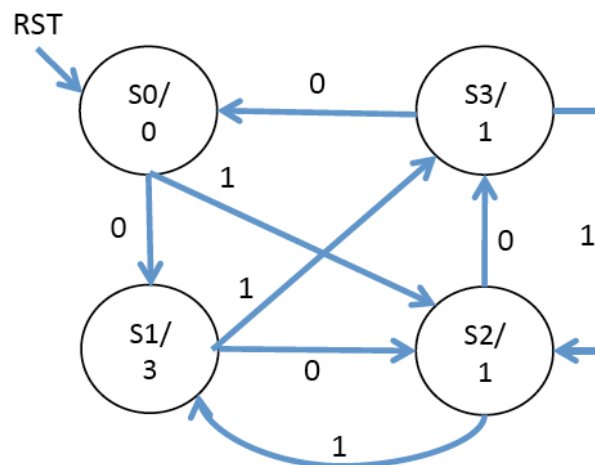


Figure 2.2

3. Figure 3.1a shows the block diagram of a circuit that produces a pair of sinusoidal signals exactly $\pi/2$ radians apart (they are known as quadrature signals). The circuit consists of a dual-port 1024 x 9-bit ROM, a 10-bit binary counter, a binary adder and a clock divider circuit as shown. The clock divider divides the 50MHz system clock by a factor of K. The binary adder adds a constant value N to the counter output cnt[9:0]. The ROM stores the coefficient values for one complete cycle of a sinewave in 2's complement form and has a Verilog HDL interface shown in Figure 3.1b.

a) Explain how this circuit works.

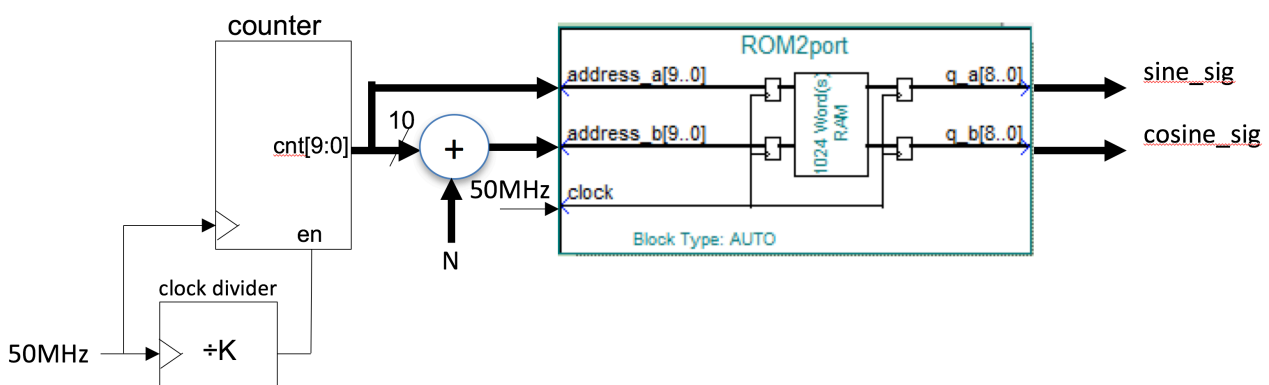
[10]

b) If you are required to produce the sine and cosine signals at a frequency closest to 10kHz, determine the value of N and K.

[5]

c) Implement in Verilog HDL, the circuit shown in Figure 2.1a. State to 4 significant digits the frequency of the output signals.

[10]



(a)

```

module ROM2port (
    address_a,
    address_b,
    clock,
    q_a,
    q_b);

    input [9:0] address_a;
    input [9:0] address_b;
    input clock;
    output [8:0] q_a;
    output [8:0] q_b;

```

(b)

Figure 3